Appearance Manifold User Guide

Using Weathering Manager Component for runtime weathered textures

1. Purpose

Appearance Manifold turns weathered texture states produced by GammaTon into runtime-ready intermediate textures. This guide is for artists and developers who want to place the Weathering Manager Component on an actor, save keyframes, generate a trajectory, and play the weathering progression in game.

> Core idea: GammaTon exports weathered BaseColor, Specular, and Roughness images. Appearance Manifold stores them as keyframe tensors, generates a trajectory, and interpolates between keyframes at runtime.

2. Prerequisites

- The target actor must have a MeshComponent. The current Blueprint automatically finds the owner's MeshComponent.
- The tested Blueprint applies the material at Material Index 0.
- Texture resolution should match Sample Size and GammaTon Texture Size. The test video used 512 x 512.
- GammaTon simulation images are loaded from Saved/GammaTon/Manifold.

3. Material Requirements

The runtime node updates texture parameters on a Dynamic Material Instance. The target material must expose these Texture Object Parameters.

Parameter	Connection
Weathering_Base	Runtime BaseColor texture. Connect it to Base Color.
Weathering_Spec	Runtime Specular texture. Connect it to Specular.
Weathering_Rough	Runtime Roughness texture. Connect it to Roughness.

> Example material: The test material uses three Texture Object Parameters feeding Texture Sample nodes, then connects RGB outputs to Base Color, Specular, and Roughness.

4. Component Setup

1. Add WeatheringManagerComponent to the target actor.
2. Assign Sample Base Color, Sample Specular, and Sample Roughness as the initial reference textures.
3. Set Sample Size to match the input texture resolution.
4. Enable Allow Weathering if automatic runtime playback is required.
5. Set Weathering Speed and Max Weathering Step.

Option	Meaning
Allow Weathering	Enables the BeginPlay playback path.
Weathering Speed	Timer interval. Lower values update more often and make progression faster.
Max Weathering Step	Number of global progression steps.
Key Weathered Textures Count	Number of saved keyframe tensors. It can be refreshed from files.

5. Keyframe Workflow

1. Create a desired weathered state with GammaTon and export the PNG results to Saved/GammaTon/Manifold.

2. Run Get New Key Frame on WeatheringManagerComponent.
3. The component loads the BaseColor, Specular, and Roughness PNGs as one 7D tensor.
4. The tensor is saved to Content/TensorCache/[ActorName]_[Index].bin.
5. Key Weathered Textures Count is incremented.

> Clear Key Frame Texture: This removes old TensorCache files, resets the count to 0, and registers Sample Base/Spec/Rough again as keyframe 0.

6. Generate Weathering Trajectory

Generate Weathering Trajectory uses the middle saved keyframe, Key Weathered Textures Count / 2, as the representative input. It reconstructs BaseColor, Specular, and Roughness textures from that tensor and passes them to the Python Appearance Manifold pipeline.

1. Load the middle keyframe tensor from Content/TensorCache.
2. Reconstruct three textures with Reconstruct Textures for Trajectory.
3. Resolve the working directory with Get Appearance Manifold Directory.
4. Call Run Weathering Pipeline to create the runtime trajectory binary.
5. Check Output Log for Binary Load Success and Trajectory Length messages.

7. Runtime Playback

BeginPlay first refreshes the keyframe count from files. Automatic playback starts only when Allow Weathering is true and at least two keyframes exist.

1. After a 2 second delay, load the trajectory binary.
2. Store Trajectory Point Count and Trajectory Samples.
3. Load TensorCache files for keyframes 0 and 1.
4. Call Apply Weathered Texture once for the initial runtime state.
5. Start a looping Weathering Step Up timer using Weathering Speed as the interval.

Runtime Event	Role
Weathering Step Up	Increments Weathering Step until Max Weathering Step and applies the next result.
Get Key Index	Maps the global step to a keyframe segment and Alpha.
Apply Weathered Texture	Builds the intermediate textures and assigns them to the mesh material.

8. Key Index Formula

Value	Formula
RealValue	((WeatheringStep - 1) * (KeyWeatheredTexturesCount - 1) / MaxWeatheringStep) + 1
IndexValue	Truncate(RealValue)
Alpha	RealValue - IndexValue

Important: Set Key Frame Count With Files counts contiguous files from index 0. If index 2 is missing, later files such as index 3 may be ignored.

9. Apply Event Internals

1. BuildNearestTrajectoryCacheAsync runs for Tex A 7D and Tex B 7D.
2. After both async caches complete, InterpolateWeatheringCached computes the intermediate 7D tensor.
3. ReconstructTexturesFromTensor creates or reuses BaseColor, Specular, and Roughness UTexture2D objects.
4. If the owner's MeshComponent is valid, ApplyWeatheringTexturesToMesh updates the material parameters.
5. The generated Dynamic Material Instance is stored in Out MID and reused on later calls.

10. Troubleshooting

Symptom	Check
Mesh does not change	Check that material parameter names match Weathering_Base, Weathering_Spec, and Weathering_Rough.

Symptom	Check
Playback does not start	Check Allow Weathering and confirm that at least two TensorCache keyframes exist.
File not found	Check Saved/GammaTon/Manifold, Content/TensorCache, and Content/WeatheringResults.
Speed feels wrong	Weathering Speed is a timer interval, not an intensity value.
A keyframe is skipped	Ensure TensorCache file indices are contiguous from 0.